

A SEMINAR REPORT ON
GENERATIVE AGENTS IN AGENT-BASED MODELING:
OVERVIEW, VALIDATION, AND EMERGING
CHALLENGES

Submitted By
ARAVIND A KAMATH (CEC23CS041)

under the esteemed guidance of
Mrs. JAYASREE K
(Assistant Professor)
Department of Computer Science and Engineering



SEPTEMBER 2026
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
COLLEGE OF ENGINEERING, PALLIPPURAM P O,
CHERTHALA,
ALAPPUZHA PIN: 688541,
PHONE: 0478 2553416, FAX: 0478 2552714
<http://www.cectl.ac.in>

A SEMINAR REPORT ON
GENERATIVE AGENTS IN AGENT-BASED MODELING:
OVERVIEW, VALIDATION, AND EMERGING
CHALLENGES

Submitted By
ARAVIND A KAMATH (CEC23CS041)

under the esteemed guidance of

Mrs. JAYASREE K

(Assistant Professor)

In partial fulfillment of the requirements for the award of the degree
of

Bachelor of Technology

in

Computer Science and Engineering

of

APJ Abdul Kalam Technological University



SEPTEMBER 2026
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
COLLEGE OF ENGINEERING, PALLIPPURAM P O,
CHERTHALA,
ALAPPUZHA PIN: 688541,
PHONE: 0478 2553416, FAX: 0478 2552714
<http://www.cectl.ac.in>

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
COLLEGE OF ENGINEERING CHERTHALA
ALAPPUZHA-688541**



C E R T I F I C A T E

This is to certify that, the seminar report titled ***GENERATIVE AGENTS IN AGENT-BASED MODELING: OVERVIEW, VALIDATION AND EMERGING CHALLENGES*** is a bonafide record of the **CSQ413 SEMINAR** presented by **ARAVIND A KAMATH (CEC23CS041)**, Seventh Semester B.Tech. Computer Science & Engineering student, under our guidance and supervision, in partial fulfillment of the requirements for the award of the degree B.Tech. Computer Science & Engineering of APJ Abdul Kalam Technological University.

Guide

Co-ordinator

HoD

Mrs. Jayasree K

Mrs. Deepa C G

Dr. Preetha Theresa Joy

Assistant Professor

Assistant Professor

Professor

Dept. of Computer Science
and Engineering

Dept. of Computer Science
and Engineering

Dept. of Computer Science
and Engineering

ACKNOWLEDGEMENT

This seminar work would not have been possible without the guidance and support of many individuals. First and foremost, I express my gratitude to Almighty God for providing the perseverance and strength to complete this seminar successfully.

I express my sincere thanks to our Principal, **Dr. Jaya V L**, for providing institutional facilities and a conducive academic environment.

I am deeply grateful to our Head of Department, **Dr. Preetha Theresa Joy**, for her encouragement and support throughout this work.

I extend my appreciation to my seminar coordinators, **Mrs. Deepa C G**, Assistant Professor, Department of Computer Science and Engineering, **Mrs. Seethamol S**, Assistant Professor, Department of Computer Science and Engineering and **Mrs. Vidya S Mony**, Assistant Professor, Department of Computer Science and Engineering for their coordination, timely reviews and constructive guidance.

I express my deepest gratitude to my guide, **Mrs. JAYASREE K**, Assistant Professor, Department of Computer Science and Engineering, for her continuous mentoring, valuable insights and guidance at every stage of this seminar.

Finally, I thank my friends and classmates for their encouragement and support during this work.

ABSTRACT

Agent-Based Modeling (ABM) is an established bottom-up computational paradigm for analyzing complex adaptive systems across epidemiology, economics and urban planning. Classical ABMs rely primarily on hand-coded heuristics or reinforcement learning policies governed by neoclassical assumptions of perfect economic rationality (*homo economicus*). Consequently, traditional models struggle to capture cognitive heterogeneity, bounded rationality, psychological nuance and natural language communication. The emergence of Large Language Models (LLMs) and Generative Artificial Intelligence has enabled a transformative paradigm shift through the development of Generative Agents (GAs) and Generative Agent-Based Models (GABMs). By embedding LLMs within the decision loop of simulation entities, GABMs enable virtual individuals capable of natural language interaction, episodic memory retention, reflection and recursive planning.

This seminar report presents an in-depth investigation of the survey paper titled “Generative Agents in Agent-Based Modeling: Overview, Validation and Emerging Challenges” published by Adornetto et al. in *IEEE Transactions on Artificial Intelligence* (December 2025). The report clarifies the distinction between Joshua Epstein’s classic generative social science paradigm and modern LLM-driven generative simulations. It details the core cognitive modules of generative agents, including the memory stream retrieval formula balancing recency, importance and relevance, hierarchical reflection trees and recursive planning mechanisms. Prominent implementations such as Smallville, Concordia, SOTOPIA, S3 and the CRSEC normative pipeline are systematically examined. Crucially, the report investigates the fundamental validation dilemma posed by transitioning from transparent white-box models to opaque, probabilistic black-box LLMs, formalizing an integrated validation blueprint combining empirical data analytics, face validation and automated LLM-as-a-Judge protocols. Finally, critical emerging challenges including hallucinations and computational complexity are analyzed, culminating in the proposed hybrid ABM-GABM architecture that couples deterministic physics engines with high-level cognitive language models to achieve both computational scalability and behavioral authenticity.

CONTENTS

1	INTRODUCTION	1
1.1	Background and Scope of Agent-Based Modeling	1
1.2	Limitations of Classical Behavioral Models	1
1.3	The Rise of Generative AI and Generative Agents	2
1.4	Seminar Objectives and Report Organization	2
2	ABMS AND GENERATIVE ABMS	3
2.1	Definitions and Mathematical Foundations	3
2.2	Generative Social Science vs. Generative Artificial Intelligence	3
2.3	Classical Generative Models: Schelling and Boids	4
2.4	Modeling Purposes and Urban Traffic Flow Case Study	4
3	EVOLVING AGENTS	6
3.1	Historical Trajectory of Agent Decision Architectures	6
3.2	Evolutionary Algorithms in Agent Systems	6
3.3	Reinforcement Learning in Multi-Agent Environments	6
3.4	Generative AI Integration in ABM Development	7
4	GENERATIVE AGENTS	9
4.1	Formal Architectural Definition	9
4.2	Parameters and Status Representation	9
4.3	Decision Architectures: Inferential vs. Conversational Loops	9
4.4	Core Cognitive Modules	10
4.5	Existing GABM Implementations	11
5	INFORMAL THEORY OF ABMS VALIDATION	12
5.1	Epistemological Principles of Simulation Validation	12
5.2	Foundational Validation Methodologies	12
5.3	Advanced Analytical Validation Methods	13
5.4	Empirical and Face Validation Dimensions	13
6	GABMS VALIDATION	14
6.1	The Fundamental Validation Dilemma	14
6.2	GABM Validation Workflows	14
6.3	Validating Existing Generative Agent Implementations	15
6.4	Automated Validation: LLM-as-a-Judge and Algorithmic Fidelity	15

7	ABM VS GABM: A COMPARISON	16
7.1	Systematic Comparative Dimensions	16
7.2	Key Trade-Offs and Architectural Insights	17
7.3	Complementarity and Methodological Synthesis	17
8	GABMS: EMERGING CHALLENGES	18
8.1	Challenge 1: Large Language Model Hallucinations	18
8.2	Challenge 2: Validation Bottlenecks and Nuance Detection	18
8.3	Challenge 3: Computational Overhead and Scaling Bottlenecks	18
9	DISCUSSION	20
9.1	Addressing Broader Epistemological Questions	20
9.2	Cities: A Natural Testbed for GABMs	20
9.3	The Proposed Paradigm: Combined Hybrid ABM-GABM Architecture	20
10	CONCLUSION	22
10.1	Summary of Key Technical Insights	22
10.2	Future Research Trajectories	23
	References	24

LIST OF FIGURES

1	AI integration across the two distinct areas of ABM development: Structural Specification and Output Analysis	7
2	LLM-based feedback loop and status of Generative Agents: (a) Inferential GA feedback loop and (b) Conversational GA feedback loop	10
3	Validation workflows for GABMs, illustrating how various validation methods align with distinct validation purposes	14

LIST OF TABLES

1	Comparison between traditional ABM and Generative ABM (Urban Traffic Flow Scenario).....	5
2	Comparison of existing GABMs, highlighting scenarios, agent characteristics and validation methods	11
3	Comparison between traditional ABM paradigms and Generative ABM (GABM) ..	16

Chapter 1

INTRODUCTION

1.1 Background and Scope of Agent-Based Modeling

Agent-Based Modeling (ABM) is an established bottom-up computational paradigm designed to simulate the simultaneous actions and autonomous interactions of individual entities, designated as agents, to assess how collective systemic phenomena emerge across time [15, 16]. Unlike classical aggregate top-down modeling techniques that rely on systems of differential equations or mean-field approximations, ABMs operate from the micro-level upward. By explicitly instantiating autonomous agents endowed with distinct internal states, behavioral heuristics, localized perception scopes and decentralized communication channels, researchers observe how macro-level regularities spontaneously emerge from localized micro-level interactions.

Over several decades, ABMs have become foundational across multiple disciplines [14]. In epidemiology, ABMs model pathogen transmission through contact networks to evaluate targeted non-pharmaceutical interventions. In computational finance, multi-agent platforms capture market crashes, asset bubbles and liquidity contagion. In urban planning, agent simulations evaluate vehicular routing, pedestrian dynamics and infrastructural transit modifications. The central epistemological strength of ABM lies in its ability to capture non-linear systemic dynamics that cannot be resolved through closed-form mathematical equations.

1.2 Limitations of Classical Behavioral Models

Despite their utility, traditional ABMs have historically relied on rigid mathematical formulas, discrete state transitions or rule sets built upon neoclassical economic assumptions of perfect rationality (*homo economicus*). These frameworks assume agents possess complete information, consistent preferences and unbounded computational capacity to maximize expected utility. Real-world human decision-making, however, is characterized by bounded rationality, psychological heuristics, cognitive biases, shifting contexts and emotional volatility.

Classical agent architectures exhibit several critical constraints:

1. **Cognitive Homogeneity and Coarse Heuristics:** Due to engineering complexity, agents are frequently modeled as homogeneous populations or partitioned into coarse archetypes, neglecting psychological diversity and cultural backgrounds.
2. **Absence of Natural Language Interaction:** Communication in traditional simula-

tions is restricted to exchanging discrete numerical values or binary flags. Agents cannot hold open-ended conversations, negotiate compromises or interpret subjective nuances.

3. **Static Memory and Lack of Cognitive Reflection:** Past interactions are either discarded or reduced to running averages. Agents lack chronological episodic memories and cannot synthesize past experiences into higher-order abstract beliefs.
4. **Brittleness and Inflexible Adaptation:** Hand-crafted if-then rules cannot adapt when agents encounter novel scenarios outside their pre-programmed logic, rendering traditional models brittle in open-ended social settings.

1.3 The Rise of Generative AI and Generative Agents

The emergence of Large Language Models (LLMs) built upon Transformer architectures [12] provides a foundation to overcome these limitations. Pretrained on massive multi-modal and textual corpora, LLMs encapsulate representations of commonsense reasoning, psychological archetypes and linguistic capabilities.

By embedding LLMs within the decision loop of simulation entities, researchers have developed **Generative Agents (GAs)** and **Generative Agent-Based Models (GABMs)** [1]. Rather than adhering to hand-crafted if-then rules, generative agents perceive environmental cues, retrieve relevant episodic memories, generate reflections, formulate daily plans and communicate via natural language dialogue. This represents an epistemological transition in computational social science: moving from mathematically bounded entities toward cognitively rich, linguistically interactive human simulacra [2].

1.4 Seminar Objectives and Report Organization

The objective of this seminar report is to provide a comprehensive technical study of the survey paper by Adornetto et al. [1] published in *IEEE Transactions on Artificial Intelligence* (December 2025). The report investigates the architectural foundations, evolution mechanisms, validation methodologies and open challenges of Generative Agents in ABM.

The report is organized into ten chapters matching the base paper: Chapter 1 introduces the background; Chapter 2 examines ABMs and Generative ABMs; Chapter 3 reviews Evolving Agents; Chapter 4 details Generative Agents; Chapter 5 outlines the Informal Theory of ABMs Validation; Chapter 6 investigates GABMs Validation; Chapter 7 provides a Comparison between ABM and GABM; Chapter 8 analyzes Emerging Challenges; Chapter 9 discusses urban testbeds and hybrid architectures; and Chapter 10 concludes the report.

Chapter 2

ABMS AND GENERATIVE ABMS

2.1 Definitions and Mathematical Foundations

An Agent-Based Model is formalized as a computational system consisting of autonomous decision-making entities situated within an environmental space. Formally, a multi-agent system at time t is defined by the tuple:

$$\mathcal{S}_t = \langle \mathcal{A}_t, \mathcal{E}_t, \mathcal{I}_t \rangle \quad (1)$$

where $\mathcal{A}_t = \{a_1, a_2, \dots, a_N\}$ denotes the set of N autonomous agents, $\mathcal{E}_t$ represents the environmental state space and $\mathcal{I}_t \subseteq \mathcal{A}_t \times \mathcal{A}_t$ specifies the interaction topology connecting agents.

Each agent a_i maintains an internal state $s_{i,t}$, sensory perception functions $P_i : \mathcal{E}_t \rightarrow \mathcal{O}_{i,t}$ and a behavioral policy π_i determining its action $u_{i,t}$:

$$u_{i,t} = \pi_i(s_{i,t}, \mathcal{O}_{i,t}) \quad (2)$$

State transitions evolve dynamically according to:

$$s_{i,t+1} = f_s(s_{i,t}, u_{i,t}, \mathcal{O}_{i,t}), \quad \mathcal{E}_{t+1} = f_e(\mathcal{E}_t, u_{1,t}, \dots, u_{N,t}) \quad (3)$$

In traditional ABMs, π_i is explicitly defined via mathematical functions or heuristics. Systemic macro-properties emerge from localized micro-actions without being explicitly programmed.

2.2 Generative Social Science vs. Generative Artificial Intelligence

The base research paper resolves the conceptual ambiguity surrounding the term “generative” in computational science by distinguishing two historically separated definitions:

1. **Generative Social Science (Epistemological):** Formulated by Joshua Epstein (1999) [4] through the motto “*If you didn’t grow it, you didn’t explain it.*” A social phenomenon is explained if and only if it can be computationally generated by decentralized, locally interacting agents without central coordination. In this framing, agent rules are typically simple; “generative” pertains strictly to the emergence of macro-patterns from micro-components.
2. **Generative Artificial Intelligence (Technological):** Refers to machine learning

models (e.g., Transformers, GANs, Diffusion Models) trained on massive corpora to generate novel high-dimensional content mirroring human artifacts.

In **Generative Agent-Based Modeling (GABM)**, these paradigms converge: micro-level decision-making is powered by Generative AI, enabling the system to be generative at both the individual cognitive level and the collective macroscopic level.

2.3 Classical Generative Models: Schelling and Boids

To demonstrate the foundational power of generative social science without generative AI, the base paper reviews two seminal historical models:

1. **Schelling's Segregation Model (1971) [3]:** Demonstrates how mild individual neighbor preferences generate severe macro-level residential segregation. In a spatial grid, agents relocate if the proportion of similar neighbors falls below a tolerance threshold. The interaction dynamics trigger spatial tipping points, producing starkly segregated neighborhoods despite mild preferences.
2. **Reynolds' Boids Model (1987) [5]:** Coordinated flocking emerges from three local steering rules: Separation (avoid crowding flockmates), Alignment (match velocity with flockmates) and Cohesion (move toward flock center). Lifelike collective behavior emerges spontaneously without central control.

2.4 Modeling Purposes and Urban Traffic Flow Case Study

The base paper adopts the seven modeling purposes formalized by Edmonds et al.: Explanation, Prediction, Description, Theoretical Exposition, Illustration, Analogy and Social Learning. Models designed for **Explanation**, **Theoretical Exposition** or **Social Learning** are inherently generative of social science because they explain phenomena through emergent internal mechanisms.

To illustrate this distinction, the base paper compares a municipal traffic simulation under both paradigms, summarized in Table 1.

Table 1: Comparison between traditional ABM and Generative ABM (Urban Traffic Flow Scenario)

Dimension	Traditional ABM	Generative ABM (GABM)
Objective	Optimizes vehicular throughput, predicts congestion bottlenecks and visually displays flow. Used as an engineering optimization tool without advancing social science theory.	Analyzes how social norms, personal risk attitudes, stress, environmental values and collective behavior shape urban mobility patterns and societal outcomes.
Outcomes	Generates context-specific engineering outputs (e.g., traffic signal timings). Findings do not transfer to other urban contexts or broader social theories.	Generates theoretical insights into the interplay between social behavior and infrastructure. Findings generalize across urban systems and collective action dilemmas.
Decision Basis	Deterministic shortest-path heuristics (Dijkstra, A*) and fixed congestion penalties, ignoring psychological states.	Context-sensitive reasoning accounting for heterogeneous values, dynamic schedules, comfort, ethics and interpersonal dialogue.

Chapter 3

EVOLVING AGENTS

3.1 Historical Trajectory of Agent Decision Architectures

Agent decision architectures have evolved from deterministic Finite State Machines (FSMs) toward adaptive optimization models. While FSMs provide computational efficiency ($O(1)$ time complexity), expanding state spaces creates combinatorial explosion, making hand-crafted rules unmaintainable in complex environments.

3.2 Evolutionary Algorithms in Agent Systems

To incorporate adaptation, researchers introduced **Evolutionary Algorithms (EAs)** grounded in Neo-Darwinian mechanics [6]. Agent policies are encoded as chromosomes and optimized through:

1. **Selection:** Allocating reproduction opportunities based on objective fitness;
2. **Crossover:** Merging parameters from high-performing parent policies; and
3. **Mutation:** Introducing random parameter perturbations to explore search spaces.

Advanced variants include **Evolution Strategies (ES)** and **Cultural Algorithms (CAs)**, which augment populations with a shared *Belief Space* storing normative, spatial and domain knowledge. However, EA models suffer from premature convergence on local optima, sensitivity to hyperparameter tuning and lack of semantic interpretability. In traffic simulations, premature convergence can cause all agents to avoid a specific road, inadvertently creating severe bottlenecks elsewhere [1].

3.3 Reinforcement Learning in Multi-Agent Environments

The integration of **Reinforcement Learning (RL)** [7] frames decision-making as a Markov Decision Process $\langle \mathcal{S}, \mathcal{U}, \mathcal{P}, \mathcal{R}, \gamma \rangle$, where agents maximize expected discounted rewards:

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{k=0}^{\infty} \gamma^k \mathcal{R}(s_{t+k}, u_{t+k}) \right] \quad (1)$$

In multi-agent settings, RL has been applied to pedestrian dynamics and traffic management. In pedestrian mobility simulations, agents optimize routes by receiving positive rewards for exit progress and penalties for collisions. However, RL relies on hand-crafted reward functions that require domain engineering. Furthermore, rigid formulations

limit adaptability beyond programmed tasks, lacking the dynamic reasoning needed for negotiation, deception and persuasion.

3.4 Generative AI Integration in ABM Development

Generative AI addresses these challenges by contributing to two distinct areas of ABM development: **Structural Specification** and **Output Analysis**, illustrated in Figure 1.

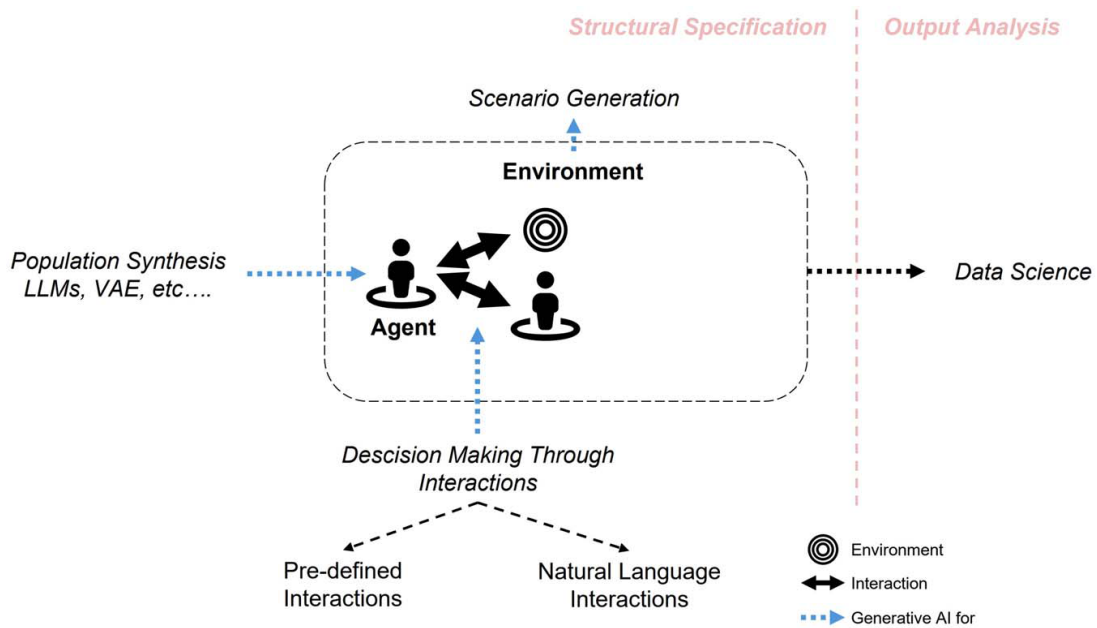


Figure 1: AI integration across the two distinct areas of ABM development: Structural Specification and Output Analysis (Source: Base Research Paper Figure 1)

3.4.1 Structural Specification

- **Population Synthesis:** LLMs synthesize representative populations endowed with realistic demographics and Big Five personality profiles based on census data.
- **Decision Making via Interactions:** Foundation models act as decentralized cognitive engines. Agents interpret situations and negotiate via natural language dialogue.
- **Scenario Generation:** Generative AI synthesizes dynamic environmental shocks, resource crises and policy interventions without manual scripting.

3.4.2 Output Analysis

- **Macro-Level Emergence Tracking:** Deep learning models track high-dimensional simulation state logs to detect subtle emergent social patterns and tipping points.

- **Narrative Summarization and Synthetic Interviewing:** Observers can directly interview individual agents or synthetic communities in natural language regarding the rationale behind their choices.

Chapter 4

GENERATIVE AGENTS

4.1 Formal Architectural Definition

A Generative Agent is defined as an autonomous computational entity capable of:

1. Perceiving multimodal or natural language sensory streams from its environment;
2. Maintaining an evolving chronological memory stream of past experiences;
3. Generating reflective higher-order inferences and abstract beliefs;
4. Devising and dynamically revising hierarchical plans; and
5. Communicating with other agents through fluent natural language dialogue.

Architecturally, GABMs decouple simulations into a **Mechanical Environment Layer** (managing spatial coordinates, collision physics and clock time) and a **Cognitive Layer** (managing LLM-driven reasoning, memory and dialogue).

4.2 Parameters and Status Representation

The base research paper establishes a clear distinction between an agent’s static parameters and dynamic status:

- **Static Baseline Parameters:** Invariant identity attributes initialized prior to simulation, including demographic data, baseline Big Five personality traits and static social network graphs. These are persistent prompts anchoring agent identity.
- **Dynamic Runtime Status:** Fluctuating state variables that evolve during simulation, such as emotional affect, spatial coordinates, active goals and chronological memories. These are maintained externally in structured text decoupled from the prompt engine.

4.3 Decision Architectures: Inferential vs. Conversational Loops

The cognitive loop operates via two distinct feedback architectures, illustrated in Figure 2:

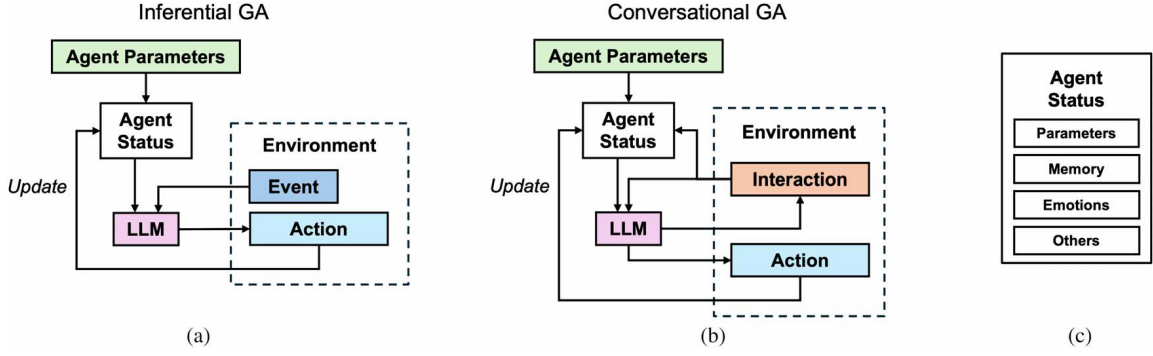


Figure 2: LLM-based feedback loop and status of Generative Agents: (a) Inferential GA feedback loop and (b) Conversational GA feedback loop (Source: Base Research Paper Figure 2)

- (a) **Inferential Feedback Loop:** As shown in Figure 2(a), the agent ingests its internal status and an environmental event trigger, executes inference and directly outputs a discrete physical action that updates the simulation environment.
- (b) **Conversational Feedback Loop:** As shown in Figure 2(b), incoming communicative events trigger an internal status update, prompting the agent to synthesize a natural language response transmitted to neighboring agents.

4.4 Core Cognitive Modules

State-of-the-art generative agents incorporate three core cognitive modules:

4.4.1 Memory Stream and Retrieval Scoring

Agents maintain a chronological database recording all perceptions and actions in natural language strings with timestamps. Because LLM context windows are bounded, relevant memories are retrieved at each step using a linear scoring function:

$$\text{Score}(m, q) = \alpha \cdot \text{Recency}(m) + \beta \cdot \text{Importance}(m) + \gamma \cdot \text{Relevance}(m, q) \quad (1)$$

where $\alpha, \beta, \gamma \geq 0$ are non-negative weighting parameters:

1. **Recency:** Decays exponentially based on elapsed simulation time: $\text{Recency}(m) = e^{-\lambda(t_{\text{current}} - t_{\text{last_access}})}$ with decay parameter $\lambda > 0$.
2. **Importance:** Integer rating from 1 (mundane) to 10 (critical), assigned by the LLM upon memory creation to distinguish trivial events from pivotal occurrences.
3. **Relevance:** Cosine similarity between dense embedding vectors of the memory record $\mathbf{e}_m$ and situational query $\mathbf{e}_q$: $\text{Relevance}(m, q) = \frac{\mathbf{e}_m \cdot \mathbf{e}_q}{\|\mathbf{e}_m\| \|\mathbf{e}_q\|}$.

4.4.2 Hierarchical Reflection Engine

To prevent retrieval from becoming flooded with granular observations, the reflection engine periodically synthesizes raw memories into abstract insights. Triggered when the cumulative importance of recent memories exceeds a threshold, the agent formulates salient high-level questions and synthesizes insights that are stored back in the memory stream, creating a hierarchical reflection tree.

4.4.3 Dynamic Recursive Planning

Planning operates recursively: agents draft a broad daily agenda, decompose hourly tasks into 5-to-15 minute action sequences and monitor environmental feedback. If unexpected events occur, execution is interrupted to revise plans dynamically.

4.5 Existing GABM Implementations

Table 2 summarizes prominent GABM platforms analyzed in the base paper.

Table 2: Comparison of existing GABMs, highlighting scenarios, agent characteristics and validation methods

Ref	Year	Framework & Scenario	Contribution Highlights	Validation Methodology
[8]	2024	SOTOPIA: Goal-driven social interactions (cooperation, competition).	Multi-agent benchmark with diverse persona profiles.	Human evaluation vs. LLM-as-a-judge across 7 dimensions.
[9]	2024	CRSEC: Social norm emergence in Smallville using 10 GAs.	4-stage pipeline: Creation, Spreading, Evaluation, Compliance.	30 human evaluators testing conflict reduction and norm adoption.
[10]	2023	Concordia: Grounded physical and social space.	Modular language agents with centralized Game Master.	Qualitative analysis and design pattern guidelines.
[2]	2023	Smallville: 25 agents in 2D RPG town forming relationships.	Architecture integrating memory stream, reflection, planning.	100 human evaluators; interview believability scoring.
[11]	2023	S3 Platform: Social network information diffusion.	LLM agents with attitude and emotion tracking.	Empirical matching against real-world social network traces.

Chapter 5

INFORMAL THEORY OF ABMS VALIDATION

5.1 Epistemological Principles of Simulation Validation

Validation in computational modeling establishes whether a model accurately represents the target system [17]. Traditional literature separates:

1. **Verification:** Ensuring internal correctness of code and algorithms (*“Was the model built right?”*); and
2. **Validation:** Ensuring model outputs conform to empirical real-world behavior (*“Was the right model built?”*).

Validation should be guided by a formal hierarchy of evidence, where the strongest evidence comes from ecological validity and rigorous experiments, while observational data and consistency with previous theories are weaker. The rigor required depends on model purpose: exploratory research models can operate under lower thresholds, whereas models guiding consequential real-world decisions demand high empirical proof. Models should also be evaluated on their productivity in stimulating further research.

5.2 Foundational Validation Methodologies

Traditional ABM validation relies on five established foundational methods:

1. **Data Analytics:** Statistical comparison of simulated outputs with historical measurements using goodness-of-fit tests, regression residuals and distribution tests.
2. **Model Docking (Alignment):** Comparing two independent simulation implementations calibrated against identical inputs to verify emergent behaviors are genuine systemic features rather than software artifacts.
3. **Empirical Validation:** Initializing simulations with historical conditions (t_0) and testing predictive accuracy against observed data up to t_1 .
4. **Sampling Techniques:** Latin Hypercube Sampling (LHS) and Monte Carlo sweeps to evaluate stability across high-dimensional parameter spaces.
5. **Visualization:** Graphical displays to detect anomalies in emergent spatial and temporal patterns.

5.3 Advanced Analytical Validation Methods

Advanced validation frameworks incorporate:

- **Bootstrapping:** Non-parametric resampling to establish parameter confidence intervals without restrictive distribution assumptions.
- **Causal Analysis:** Directed Acyclic Graphs (DAGs) and counterfactual intervention testing to differentiate genuine causal mechanisms from statistical correlations.
- **Inverse Generative Social Science (IGSS):** Using genetic programming to search the space of executable agent rules that generate target macroscopic phenomena.
- **Spatial Analytics:** Moran's I and spatial point pattern statistics to validate simulated spatial distributions against empirical GIS data.

5.4 Empirical and Face Validation Dimensions

In social simulations where quantitative data is noisy, researchers rely on **Face Validation** to assess behavioral credibility:

1. **Microface Validation:** Evaluates individual agent choices against established psychological and behavioral theories.
2. **Macroface Validation:** Assesses whether aggregate population distributions exhibit realistic macroscopic structures.
3. **Face Validation Methods:** Includes Animation Assessment (visual tracking of agent behavior), Output Assessment (expert plausibility audits) and Immersive Assessment (validators impersonate agents via interactive interfaces or virtual reality to evaluate emotional dynamics firsthand).

Chapter 6

GABMS VALIDATION

6.1 The Fundamental Validation Dilemma

The integration of LLMs introduces the **Fundamental Validation Dilemma**. While traditional ABMs are transparent white-box systems with deterministic state transitions, Generative Agents rely on foundation models that function as probabilistic black boxes:

1. **Stochastic Non-Determinism:** Temperature-based decoding generates divergent paths from identical starting seeds, invalidating deterministic replication.
2. **Opaque Internal Reasoning:** Internal states are distributed across billions of parameters without explicit algebraic transition equations.
3. **High-Dimensional Output Spaces:** Agent actions include unconstrained natural language requiring evaluation of semantic coherence, emotional nuance and truthfulness.

6.2 GABM Validation Workflows

Figure 3 illustrates the integrated validation architecture formulated in the base paper, connecting validation methods with distinct validation purposes.

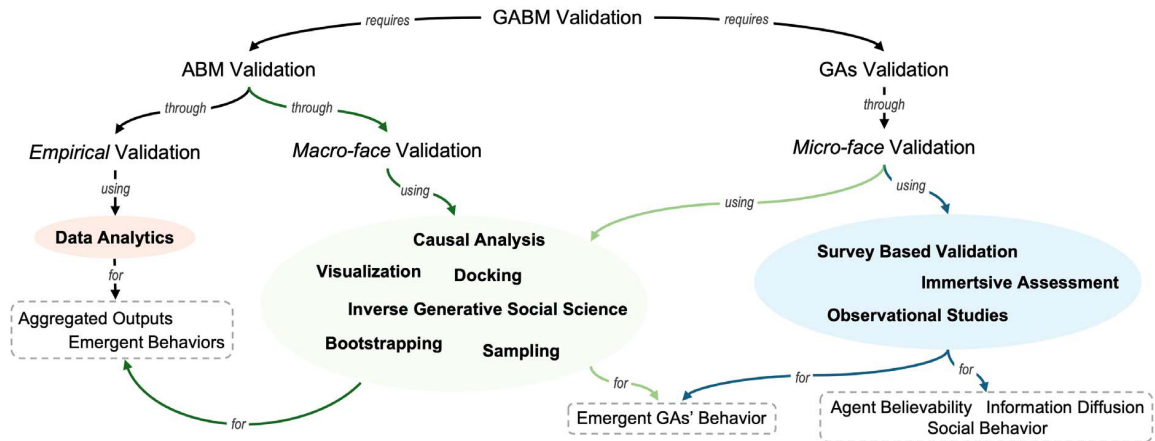


Figure 3: Validation workflows for GABMs, illustrating how various validation methods align with distinct validation purposes (Source: Base Research Paper Figure 3)

The framework operates across three distinct pathways given below.

1. **Empirical Pathway:** Uses micro-level data (census, demographics) to condition baseline parameters and macro-level data (voting outcomes, traffic indices) to evaluate aggregate simulation trajectories statistically.
2. **Macroface Pathway:** Validates collective emergent phenomena (social network topologies, norm adoption, spatial clustering) arising without centralized control.
3. **Microface Pathway:** Assesses individual agent plausibility via structured interviews, ablation testing and automated dialogue audits.

6.3 Validating Existing Generative Agent Implementations

In Smallville [2], 100 evaluators interviewed agents across five categories: self-knowledge, memory retrieval, planning consistency, reaction plausibility and reflection depth. Ablation experiments demonstrated that removing memory causes repetitive behaviors, removing reflection hinders abstract reasoning and removing planning induces erratic actions; the full architecture achieved statistically superior believability.

In SOTOPIA [8], the SOTOPIA-EVAL benchmark evaluates goal-driven social interactions across seven dimensions: Goal Completion, Believability, Knowledge Gained, Secret Keeping, Relationship Dynamics, Social Norm Adherence and Financial Benefit.

6.4 Automated Validation: LLM-as-a-Judge and Algorithmic Fidelity

To overcome the cost of human evaluation, researchers employ **LLM-as-a-Judge** [20], prompting models like GPT-4 to grade multi-agent dialogues. Dual-evaluation studies indicate high correlation on objective metrics (Goal Completion, Financial Benefit), but limitations in detecting subtle social nuance, sarcasm and strategic deception.

Consequently, the base paper emphasizes **Algorithmic Fidelity** [13] and the **Similarity Principle**. Before deploying LLMs as simulation agents or evaluators, researchers must benchmark their baseline responses against validated sociological surveys [19] to detect and mitigate systemic cultural biases.

Chapter 7

ABM VS GABM: A COMPARISON

7.1 Systematic Comparative Dimensions

Table 3: Comparison between traditional ABM paradigms and Generative ABM (GABM)

Evaluation Dimension	Rule-based	EA-based	RL-based	GABM
1. Resources				
Computational Efficiency	Advantage	Limitation	Limitation	Limitation
Scalability	Advantage	Limitation	Limitation	Limitation
Data or Domain-Specific Knowledge	Advantage	Advantage	Advantage	Advantage
2. Agent Design				
Control over Agents	Advantage	Limitation	Limitation	Limitation
Learning During Simulation	Limitation	Advantage	Advantage	Limitation
Real-Time Optimization	Limitation	Advantage	Advantage	Limitation
Hyperparameters Tuning	Advantage	Limitation	Limitation	Limitation
Fast Prototyping	Limitation	Limitation	Limitation	Advantage
3. Agent Behavior				
Determinism	Advantage	Limitation	Limitation	Limitation
Reliable Decision-Making	Advantage	Limitation	Limitation	Limitation
Evolution	Limitation	Advantage	Advantage	Limitation
Cognitive Functions Simulation	Limitation	Limitation	Limitation	Advantage
Language Human-like Interactions	Limitation	Limitation	Limitation	Advantage
Flexible and Reactive Decisions	Limitation	Limitation	Advantage	Advantage
Nuanced Mental Models	Limitation	Limitation	Limitation	Advantage
4. Validation				
Transparency of Decision-Making	Advantage	Limitation	Limitation	Limitation
Validation Efficiency	Advantage	Limitation	Limitation	Limitation
Direct Human-Agent Interaction	Limitation	Limitation	Limitation	Advantage

Table 3 reproduces the comparative analysis from Table III of the base paper, contrasting traditional ABMs (Rule-based, EA-based, RL-based) with GABMs across Resources, Agent Design, Agent Behavior and Validation.

7.2 Key Trade-Offs and Architectural Insights

7.2.1 Resources: Efficiency, Scalability and Knowledge

Rule-based models require simple CPU arithmetic ($O(1)$ time complexity), scaling to millions of agents in macroeconomic or epidemic models. GABMs incur significant GPU inference latency and token costs, restricting cohorts to dozens or hundreds of agents. However, GABMs leverage broad pretrained world knowledge, eliminating the need to construct explicit domain rules from scratch.

7.2.2 Agent Design: Control, Learning and Prototyping

Rule-based models provide deterministic control over every logic branch. GABMs rely on prompt engineering, which is susceptible to semantic drift. Furthermore, LLM weights remain frozen during simulation; agent learning is confined to in-context memory accumulation. However, GABMs provide superior **Fast Prototyping**, allowing complex personas to be defined in minutes via natural language.

7.2.3 Behavior and Validation: Cognitive Fidelity vs. Transparency

Rule-based systems offer mathematical transparency and replicability, but cannot model open-ended language, emotional states or mental models. GABMs excel in cognitive realism and introduce **Direct Human-Agent Interaction**, enabling researchers to interview agents regarding their motivations.

7.3 Complementarity and Methodological Synthesis

The core conclusion of the comparative analysis in the base paper is that traditional ABMs and GABMs are complementary rather than mutually exclusive. Traditional ABMs contribute computational efficiency, deterministic control and demographic scalability; GABMs contribute cognitive realism, linguistic communication and adaptive reasoning. Effective simulation requires hybrid methodologies that combine the advantages of both paradigms.

Chapter 8

GABMS: EMERGING CHALLENGES

8.1 Challenge 1: Large Language Model Hallucinations

Hallucinations undermine GABM validity across two primary modes:

1. **Factuality Hallucinations:** Generating assertions that contradict environmental ground truth (e.g., asserting a bridge exists across an uncrossable river).
2. **Faithfulness Hallucinations:** Violating assigned persona constraints, moral values or historical context (e.g., a pacifist initiating unprovoked violence).

Mitigation strategies operate across three stages:

- **Data Stage:** Grounding prompts via Retrieval-Augmented Generation (RAG) using verified real-time simulation facts.
- **Training Stage:** Applying sensitive neuron dropout and RLHF to enforce persona consistency and penalize behavioral drift.
- **Inference Stage:** Enforcing formal grammar constraints (e.g., JSON schema) and deploying peer verification protocols where secondary agents audit outputs.

8.2 Challenge 2: Validation Bottlenecks and Nuance Detection

Automated LLM judges exhibit critical limitations:

1. **Nuance Blindness:** Evaluators struggle with passive aggression, sarcasm and strategic deception, showing bias toward artificially polite agents.
2. **Evaluator Self-Preference Bias:** Foundation models favor text produced by models of their own architectural family.
3. **Lack of Standardized Benchmarks:** Absence of standardized multi-agent benchmarks prevents rigorous cross-study comparisons.

8.3 Challenge 3: Computational Overhead and Scaling Bottlenecks

Scaling constraints arise across three bottlenecks:

1. **Quadratic Self-Attention Overhead:** Computational complexity scales quadratically with prompt token length n [12]:

$$\text{Complexity}_{\text{Attention}} = O(n^2 \cdot d) \quad (1)$$

where d is embedding dimension. Expanding memory streams escalate inference latency.

2. **Pairwise Communication Scaling:** In a network of N interacting agents, potential dialogue channels scale quadratically:

$$\text{Complexity}_{\text{Communication}} = O(N^2) \quad (2)$$

3. **Memory Retrieval Indexing:** Vector search across M records introduces indexing complexity $O(\log M)$ to $O(M)$ per agent per tick.

Chapter 9

DISCUSSION

9.1 Addressing Broader Epistemological Questions

The base research paper examines five fundamental questions driving social simulation:

1. **Macroscopic Pattern Emergence:** How macro-patterns emerge from micro-interactions with diverse, personality-driven agents.
2. **Information and Innovation Diffusion:** How ideas spread dynamically through curious, skeptical and communicative agents.
3. **Strategic Adaptation in Dynamic Environments:** How agents modify strategies autonomously in resource-limited game settings.
4. **Social Network Influence and Polarization:** How interpersonal persuasion and conformity shape echo chambers and polarization.
5. **Linguistic and Framing Biases:** How natural language framing influences social deliberation and consensus formation.

9.2 Cities: A Natural Testbed for GABMs

Urban environments provide an ideal deployment testbed:

- **Textual Alignment:** Urban regulations, historical archives and census data align with LLM training corpora, providing rich baseline representations.
- **Qualitative Experience Mapping (QEM):** Traditional models (such as MAT-Sim [18]) simulate physical trajectories. GABMs enable tracking subjective experiences: safety perceptions in unlit alleys, park aesthetic appreciation and transit stress, informing human-centric urban design.

9.3 The Proposed Paradigm: Combined Hybrid ABM-GABM Architecture

To overcome computational bottlenecks while preserving behavioral realism, the base paper proposes an integrated **Hybrid ABM-GABM Architecture**:

1. **Mechanistic Layer (Traditional ABM):** High-frequency, deterministic processes (kinematics, collision avoidance, shortest-path routing) execute at $O(1)$ or $O(N)$ without invoking LLMs.
2. **Cognitive Layer (Generative Agents):** Foundation model inference is invoked selectively only for complex deliberations, ethical dilemmas and social dialogue.

9.3.1 Case Study: Building Fire Evacuation

In an emergency building fire evacuation:

- The traditional layer computes toxic smoke dispersion and pedestrian stairwell routing via cellular automata.
- The generative layer evaluates moral dilemmas (e.g., deciding whether to re-enter a burning floor to assist a trapped colleague versus immediate evacuation).
- Sensor data analytics validate evacuation clearance durations, while post-evacuation agent interviews validate psychological panic and altruistic behaviors.

Chapter 10

CONCLUSION

10.1 Summary of Key Technical Insights

This seminar report has conducted an in-depth technical investigation into the paradigm of Generative Agents in Agent-Based Modeling, based on the comprehensive survey paper by Adornetto et al. published in *IEEE Transactions on Artificial Intelligence* (December 2025). The introduction of foundation models into agent-based modeling marks a decisive departure from neoclassical assumptions of perfect rationality (*homo economicus*) toward heterogeneous, cognitively plausible human simulacra. By embedding large language models into agent architectures through episodic memory streams, associative retrieval scoring, hierarchical reflection trees and dynamic recursive planning, generative agents demonstrate remarkable autonomy, believable social interactions and spontaneous normative emergence.

A foundational insight emphasized throughout this work is the epistemological clarification distinguishing classical generative social science from technological generative artificial intelligence. Joshua Epstein’s seminal principle established that a social phenomenon is explained if and only if it can be grown computationally from decentralized local interactions. The presence of generative artificial intelligence is not in itself sufficient to establish generative social science; rather, true explanatory power depends on modeling the internal behavioral mechanisms that give rise to macro-level regularities. Generative Agent-Based Models unite these two paradigms, leveraging language models to empower individual agent deliberation while preserving the bottom-up explanatory ethos of agent-based modeling.

The comparative analysis across resources, agent design, agent behavior and validation establishes that traditional ABMs and GABMs represent complementary rather than mutually exclusive methodologies. Traditional rule-based and evolutionary models provide computational efficiency, deterministic reproducibility and demographic scalability to millions of agents. In contrast, GABMs provide cognitive nuance, natural language communication and rapid persona prototyping. To address the black-box validation dilemma arising from opaque neural architectures, researchers have established multi-tiered validation workflows that combine quantitative empirical data analytics, qualitative face validation, human interview audits and automated LLM-as-a-Judge evaluations calibrated by algorithmic fidelity. The combined hybrid ABM-GABM architecture bridges these methodologies, coupling high-frequency deterministic physics engines with selective cognitive language model deliberation to achieve both computational tractability and behavioral authenticity.

10.2 Future Research Trajectories

Future research in Generative Agent-Based Modeling must address several critical frontiers to enable widespread adoption in academic and policy domains. A primary imperative is the development of specialized Small Language Models (SLMs) and parameter-efficient architectures. Deploying compact, on-device models fine-tuned specifically for social reasoning will substantially reduce self-attention latency, lower hardware memory demands and eliminate dependency on costly cloud API services during large-scale simulations.

Simultaneously, the computational social science community must establish standardized, open-source simulation benchmarks and shared evaluation datasets. Similar to established benchmarks in natural language processing and computer vision, standardized multi-agent environments will enable rigorous cross-study comparisons, allowing researchers to evaluate competing cognitive architectures across uniform social tasks, normative emergence scenarios and collective action dilemmas.

Rigorous mathematical frameworks must also be established to formulate formal upper bounds on hallucination probabilities within policy-critical simulations. As generative models are increasingly applied to municipal transit planning, public health protocols and socioeconomic modeling, developing constrained decoding formalisms and multi-agent peer verification protocols will be essential to ensure synthetic agents adhere strictly to environmental ground truth and assigned demographic profiles.

Finally, the expanding cognitive capabilities of generative agents necessitate the formulation of robust ethical guidelines and governance frameworks. Researchers must institute systematic algorithmic fidelity audits to prevent the unexamined propagation of cultural, demographic and socioeconomic biases inherited from foundation model pretraining corpora. Establishing rigorous ethical boundaries will ensure that generative agent simulations serve as reliable, equitable and transparent instruments for investigating complex human societies.

References

- [1] C. Adornetto, A. Mora, K. Hu, L. I. Garcia, P. Atchade-Adelomou, G. Greco, L. A. A. Pastor and K. Larson, “Generative Agents in Agent-Based Modeling: Overview, Validation and Emerging Challenges,” *IEEE Transactions on Artificial Intelligence*, vol. 6, no. 12, pp. 3165-3184, Dec. 2025, doi: 10.1109/TAI.2025.3566362.
- [2] J. S. Park, J. C. O’Brien, C. J. Cai, M. R. Morris, P. Liang and M. S. Bernstein, “Generative Agents: Interactive Simulacra of Human Behavior,” in *Proc. 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, San Francisco, CA, USA, Oct. 2023, pp. 1-22.
- [3] T. C. Schelling, “Dynamic Models of Segregation,” *Journal of Mathematical Sociology*, vol. 1, no. 2, pp. 143-186, 1971.
- [4] J. M. Epstein, “Agent-Based Computational Models and Generative Social Science,” *Complexity*, vol. 4, no. 5, pp. 41-60, May/Jun. 1999.
- [5] C. W. Reynolds, “Flocks, Herds and Schools: A Distributed Behavioral Model,” in *Proc. 14th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)*, Anaheim, CA, USA, Aug. 1987, pp. 25-34.
- [6] J. H. Holland, *Adaptation in Natural and Artificial Systems: An Introductory Analysis with Applications to Biology, Control and Artificial Intelligence*, Cambridge, MA, USA: MIT Press, 1992.
- [7] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., Cambridge, MA, USA: MIT Press, 2018.
- [8] H. Wang, X. Guo, Z. Deng, H. Shi, H. Shen, B. Zheng, E. P. Xing and Z. Hu, “Sotopia: Interactive Evaluation for Social Intelligence in Language Agents,” in *Proc. International Conference on Learning Representations (ICLR)*, Vienna, Austria, May 2024.
- [9] S. He, H. Zheng, L. Du and W. Ding, “Emergence of Social Norms in Generative Agent Societies: Principles and Architecture,” in *Proc. 38th AAAI Conference on Artificial Intelligence (AAAI)*, Vancouver, Canada, Feb. 2024, pp. 22120-22128.
- [10] A. Vezhnevets, J. P. Agapiou, A. Aharon, R. Boning, E. A. Duenez-Guzman, J. Leibo, et al., “Generative Agent-Based Modeling with Actions Grounded in Physical, Social or Digital Space Using Concordia,” *arXiv preprint arXiv:2312.03664*, Dec. 2023.
- [11] C. Gao, X. Lan, Z. Lu, J. Mao, J. Piao, H. Wang, D. Jin and Y. Li, “S³: Social-Network Simulation System with Large Language Model-Empowered Agents,” *arXiv preprint arXiv:2307.14984*, Jul. 2023.
- [12] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser and I. Polosukhin, “Attention Is All You Need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, Long Beach, CA, USA, Dec. 2017, pp. 5998-6008.
- [13] L. P. Argyle, E. C. Busby, N. Fulda, J. R. Gubler, C. Rytting and D. Wingate, “Out of One,

- Many: Using Language Models to Simulate Human Samples,” *Political Analysis*, vol. 31, no. 3, pp. 337-351, Jul. 2023.
- [14] R. Axtell, “Why Agents? On the Varied Motivations for Agent Computing in the Social Sciences,” in *Proc. Workshop on Agent-Based Modeling: Investment, Risk and Policy*, Santa Fe Institute, Santa Fe, NM, USA, 2002.
- [15] E. Bonabeau, “Agent-Based Modeling: Methods and Techniques for Simulating Human Systems,” *Proc. National Academy of Sciences (PNAS)*, vol. 99, no. suppl. 3, pp. 7280-7287, May 2002.
- [16] C. M. Macal and M. J. North, “Tutorial on Agent-Based Modelling and Simulation,” *Journal of Simulation*, vol. 4, no. 3, pp. 151-162, Sep. 2010.
- [17] V. Grimm, U. Berger, F. Bastiansen, S. Eliassen, V. Ginot, J. Giske, et al., “A Standard Protocol for Describing Individual-Based and Agent-Based Models,” *Ecological Modelling*, vol. 198, no. 1-2, pp. 115-126, Sep. 2006.
- [18] A. Horni, K. Nagel and K. W. Axhausen, *The Multi-Agent Transport Simulation MATSim*, London, UK: Ubiquity Press, 2016.
- [19] G. Serapio-Garcia, M. Mustafa, S. Verma, M. Caron, M. Matarić and A. Faust, “Personality Traits in Large Language Models,” *arXiv preprint arXiv:2307.00184*, Jul. 2023.
- [20] L. Zheng, W. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Xing, H. Zhang, J. E. Gonzalez and I. Stoica, “Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena,” in *Advances in Neural Information Processing Systems (NeurIPS)*, New Orleans, LA, USA, Dec. 2023, pp. 46595-46623.